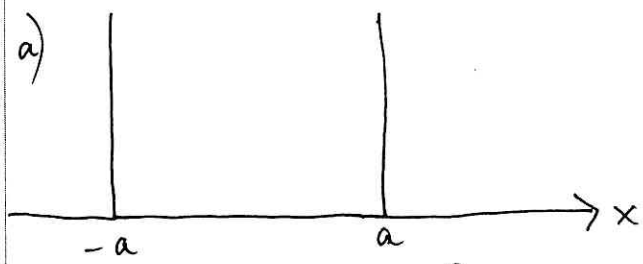
HW 04

(P1) a)



$E = \frac{mv^2}{2}$ for $|x|<a$.

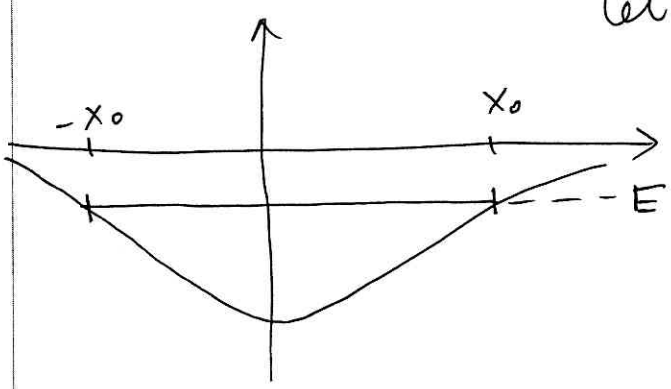
Turning points $x = \pm a$

$$T = 2 \cdot \sqrt{\frac{m}{2}} \int_{-a}^{a} \frac{dx}{\sqrt{E - U(x)}} = \sqrt{\frac{2m}{E}}\,(2a) = \sqrt{\frac{2m}{\frac{mv^2}{2}}}\,(2a) = \frac{4a}{v}$$

of course.

b) $U(x) = -\frac{U_0}{\cosh^2(\alpha x)}$, $-U_0 < E < 0$.

let $E = -|E|$.



$$E = \frac{m\dot{x}^2}{2} + U(x)$$

Turning points: $E = U(x_0)$

$$\Rightarrow |E| = \frac{U_0}{\cosh^2(\alpha x_0)}$$

$$T = 2\sqrt{\frac{m}{2}} \int_{-x_0}^{x_0} \frac{dx}{\sqrt{E + \frac{U_0}{\cosh^2(\alpha x)}}} =$$ (the integrand is even function of x) =

$$= \sqrt{2m} \cdot 2 \int_0^{x_0} \frac{dx}{\sqrt{\frac{U_0}{\cosh^2(\alpha x)} - \frac{U_0}{\cosh^2(\alpha x_0)}}} =$$

$$= 2\sqrt{\frac{2m}{U_0}} \int_0^{x_0} \frac{\cosh(\alpha x_0)\cosh(\alpha x)\,dx}{\sqrt{\cosh^2(\alpha x_0) - ch^2(\alpha x)}}$$

let $y = \sinh(\alpha x) \Rightarrow dy = \alpha \cosh(\alpha x)\,dx$; use $\cosh^2(\alpha x) = 1 + \sinh^2(\alpha x)$

$$T = \frac{2}{\alpha} \cosh(\alpha x_0) \sqrt{\frac{2m}{U_0}} \int_0^{y_0} \frac{dy}{\sqrt{\underbrace{\cosh^2(\alpha x_0) - 1}_{\sinh^2(\alpha x_0) = y_0^2} - y^2}}$$

$$y_0 = \sinh(\alpha x_0)$$

$$T = \frac{2\cosh(\alpha x_0)}{\alpha} \sqrt{\frac{2m}{U_0}} \int_0^{y_0} \frac{dy}{\sqrt{y_0^2 - y^2}} = (y = y_0 u)$$

$$= \frac{2\cosh(\alpha x_0)}{\alpha} \sqrt{\frac{2m}{U_0}} \int_0^1 \frac{y_0\,du}{y_0\sqrt{1-u^2}} = (u = \sin\varphi)$$

$$= \frac{2\cosh(\alpha x_0)}{\alpha} \sqrt{\frac{2m}{U_0}} \underbrace{\int_0^{\pi/2} \frac{\cos\varphi\,d\varphi}{\cos\varphi}}_{\frac{\pi}{2}} =$$

$$= \frac{\pi}{\alpha} \sqrt{\frac{2m}{U_0}} \cosh(\alpha x_0) = \frac{\pi}{\alpha} \sqrt{\frac{2m}{|E|}}.$$

But $\cosh(\alpha x_0) = \sqrt{\frac{U_0}{|E|}}$

HW04

P2

PROBLEMS

PROBLEM 1. Integrate the equations of motion for a *spherical pendulum* (a particle of mass m moving on the surface of a sphere of radius l in a gravitational field).

SOLUTION. In spherical co-ordinates, with the origin at the centre of the sphere and the polar axis vertically downwards, the Lagrangian of the pendulum is

$$\tfrac{1}{2}ml^2(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta)+mgl\cos\theta.$$

The co-ordinate ϕ is cyclic, and hence the generalised momentum p_ϕ, which is the same as the z-component of angular momentum, is conserved:

$$ml^2\dot{\phi}\sin^2\theta = M_z = \text{constant}. \tag{1}$$

The energy is

$$\begin{aligned} E &= \tfrac{1}{2}ml^2(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta)-mgl\cos\theta \\ &= \tfrac{1}{2}ml^2\dot{\theta}^2+\tfrac{1}{2}M_z^2/ml^2\sin^2\theta-mgl\cos\theta. \end{aligned} \tag{2}$$

Hence

$$t = \int\frac{d\theta}{\sqrt{\{2[E-U_{\text{eff}}(\theta)]/ml^2\}}}, \tag{3}$$

where the "effective potential energy" is

$$U_{\text{eff}}(\theta) = \tfrac{1}{2}M_z^2/ml^2\sin^2\theta-mgl\cos\theta.$$

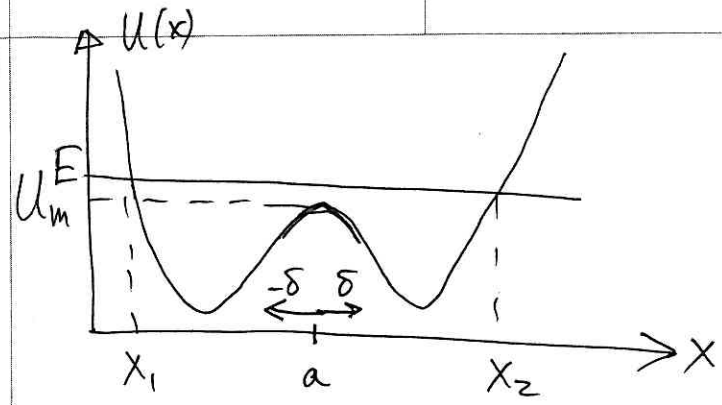
For the angle ϕ we find, using (1),

$$\phi = \frac{M_z}{l\sqrt{(2m)}}\int\frac{d\theta}{\sin^2\theta\sqrt{[E-U_{\text{eff}}(\theta)]}}. \tag{4}$$

The integrals (3) and (4) lead to elliptic integrals of the first and third kinds respectively.

The range of θ in which the motion takes place is that where $E > U_{\text{eff}}$, and its limits are given by the equation $E = U_{\text{eff}}$. This is a cubic equation for $\cos\theta$, having two roots between -1 and $+1$; these define two circles of latitude on the sphere, between which the path lies

(P3)



$E - U_m \ll U_m$

$U(x) \approx U_m - \frac{1}{2} k (x-a)^2$

Particle's speed is smallest when $x = a \Rightarrow$ it spends most time there!

$$T = 2\sqrt{\frac{m}{2}} \int_{x_1}^{x_2} \frac{dx}{\sqrt{E - U(x)}} = T_1 + T_2$$

where $T_1 = \sqrt{2m} \int_{a-\delta}^{a+\delta} \frac{dx}{\sqrt{E - U(x)}}$ is the time of motion through $(a-\delta, a+\delta)$ interval,

T_2 is the rest of the integral.

Near $x = a$ $\quad E - U(x) = E - U_m + \frac{1}{2} k (x-a)^2$

$$T_1 = \sqrt{2m} \int_{a-\delta}^{a+\delta} \frac{dx}{\left(\frac{1}{2}k\right)^{1/2} \sqrt{\frac{2(E-U_m)}{k} + (x-a)^2}} = (x = a + y)$$

$$= \sqrt{\frac{4m}{k}} \int_{-\delta}^{\delta} \frac{dy}{\sqrt{\underbrace{\frac{2(E-U_m)}{k}}_{\alpha^2} + y^2}} = 4\sqrt{\frac{m}{k}} \ln\left(y + \sqrt{\alpha^2 + y^2}\right)\Big|_0^{\delta}$$

$$= 4\sqrt{\frac{m}{k}} \ln\left(\frac{\delta + \sqrt{\alpha^2 + \delta^2}}{\alpha}\right) \to 4\sqrt{\frac{m}{k}} \ln \frac{2\delta}{\sqrt{\frac{2}{k}(E - U_m)}}$$

The integral diverges logarithmically as $E \to U_m$

$\Rightarrow T_1 \gg T_2$ then, and $T \approx T_1$.